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Remaining useful life prediction based on self-attention mechanism-sequential variational autoencoder: From a semi-supervised perspective

Jiusi Zhang^a, Kai Chen^a, Fan Wu^a, Quan Qian^a, Tenglong Huang^b, Yuhua Cheng^a, Shen Yin^c

^a School of Automation Engineering, University of Electronic Science and Technology of China, Chengdu, China

^b College of Mechanical and Electronic Engineering, Northwest A&F University, Yangling, China

^c Department of Mechanical and Industrial Engineering, Faculty of Engineering, Norwegian University of Science and Technology, Trondheim, Norway

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ABSTRACT

Remaining useful life (RUL) prediction is a crucial task in maintaining operational safety and dependability of complex systems. Considering that conventional data-driven RUL prediction approaches demand massive labeled samples for supervised model training, it has notable limitations in making full use of unlabeled degradation data. Furthermore, current deep networks require in-depth research work in terms of interpretability and uncertainty. In this sense, an RUL prediction approach with the self-attention mechanism-sequential variational autoencoder (SAM-SVAE) is proposed from a semi-supervised perspective. Specifically, considering a dynamic serialization modeling, this paper designs a self-attention mechanism network to focus on key parts of the input time window. On this basis, this paper explores the correspondence between a Bayesian deep probability generation network and the state space model in control theory, which approximates the RUL prediction's density function for uncertainty assessment. Moreover, this paper proposes an SAM-SVAE from a semi-supervised perspective, which can learn valuable feature representations from a large amount of unlabeled degradation data, from which the interpretability is provided through the analyze of latent space. Experimental validation of the presented SAM-SVAE utilizes the aircraft turbofan engine dataset from NASA Prediction Center.

1. Introduction

With the accelerating pace of smart technologies and digital transformation across industrial domains, ensuring the reliability of sophisticated systems has become fundamentally important for manufacturing productivity [1–3]. Nevertheless, the complex systems inevitably become degradation due to operational conditions, environmental changes, and material aging [4–6]. As a logical consequence, accurate prediction of remaining useful life (RUL) in complex systems represents a fundamental component of effective maintenance strategies and asset management [7–10]. Accurate RUL prediction can help users plan maintenance resource allocation. Furthermore, the unnecessary downtime and repair costs are decreased, thereby promoting production efficiency [11–13].

State-of-the-art RUL prediction can be separated into the mechanism modeling-based approaches and the data-driven approaches [14]. The mechanism modeling-based approaches often require in-depth domain knowledge to establish accurate mathematical model through the physical characteristics [15,16]. Such approaches have notable limitations

in application due to model assumptions and generalization capabilities [17,18]. In contrast, data-driven approaches utilize sensor-collected data to construct learning-based algorithmic models when prior knowledge is unavailable, which automatically mine the valuable information hidden in the degradation data [19–21]. Recently, the development of computer and information processing technology has also provided favorable conditions, which support for the data-driven RUL approaches to process massive data [22–24]. These approaches can automatically learn feature representations from data, which can better handle the temporal data generated by complex industrial systems. Committed to incorporating prior knowledge in a data-driven approach, Zhang et al. integrated attention-based recurrent neural network and particle filtering to effectively predict the RUL of lithium-ion batteries [25]. Chen et al. implemented federated learning in their pruned Taylor network to enhance data security for bearing condition-based RUL prediction [26]. Considering the changes in the degradation rate and

* Corresponding authors.

E-mail addresses: qian_1998@cqu.edu.cn (Q. Qian), huangtenglong@nwfau.edu.cn (T. Huang).

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the accumulated information, Zhang et al. proposed a two-stage data-driven approach to design a time-delay neural network to predict the RUL [27].

Considering that most complex systems are required to work in a healthy state, obtaining degradation data throughout the life cycle and corresponding RUL labels are usually expensive and time-consuming [28,29]. Conversely, conventional data-driven approaches demand massive labeled samples for supervised model training, which makes it difficult to make full use of unlabeled data. Consequently, semi-supervised learning provides an opportunity to solve the above challenges. Facing multi-objective functions, He et al. developed a semi-supervised adversarial network to predict the RUL [30]. From the perspective of global information, Zhuang et al. presented a semi-supervised dual-attention model that delivers great performance for machinery's RUL prediction [31]. Considering the uncertainty in the degradation process, Che et al. proposed a semi-supervised self-learning approach to predict the RUL of lithium-ion batteries [32]. For current semi-supervised RUL prediction methods, network performance requires further improvement when labeled data is limited. Moreover, current conventional deep learning approaches require in-depth research work in terms of interpretability and uncertainty [33–35]. Specifically, most of the deep networks are black box models. It is of great significance to explain the meaning of the latent space for the guidance of the efficient RUL prediction process. In addition, measuring the uncertainty in the RUL during the degradation process is also a vital task, which helps to promote the model credibility and practicality.

The primary motivations for this research emerge from the following factors:

- The degradation data of complex systems have strong temporal characteristics. Consequently, how to focus on key parts of the time window, and make full use of the temporal information requires further research [25,36].
- Conventional deep networks are usually regarded as black box model, the meaning of whose internal latent spaces is difficult to interpret. Furthermore, most of the RUL prediction approaches are point estimation, which requires in-depth work on uncertainty description [33–35,37].
- Conventional data-driven approaches demand massive labeled samples for supervised model training. How to effectively utilize unlabeled degradation data from a semi-supervised perspective requires further work [30–32].

Based on these observations, an RUL prediction approach based on the self-attention mechanism-sequential variational autoencoder (SAM-SVAE) is proposed from a semi-supervised perspective to predict the RUL of complex systems. The main contributions can be summarized as the follows:

- **Dynamic serialization with self-attention mechanism for degradation modeling:** A self-attention mechanism with dynamic serialization module, coupled with a bidirectional gated recurrent unit (BiGRU) encoder is proposed. The module concentrates modeling capacity on salient time steps, while preserving long-short range dependencies in multivariate sensor streams. During training and inference, it can achieve more powerful temporal feature extraction for the RUL prediction.
- **Control-theoretic latent state space with uncertainty description:** An SAM-SVAE approach induces a learned hidden-variable state space that is explicitly aligned with a state-space model from control theory. The inference sub-network, encoding sub-network, and decoding sub-network jointly define a full predictive RUL distribution, enabling calibrated uncertainty quantification and providing an interpretable latent trajectory consistent with control-theoretic dynamics.

- **Semi-supervised training that scales with limited labels:** A semi-supervised training objective jointly optimizes a supervised likelihood with an unsupervised feature reconstruction, which allows effective use of large unlabeled degradation datasets without synthetic or pseudo-labels. Experiments on the NASA's turbofan engine dataset demonstrate reduced label dependence, and excellent RUL prediction performance.

2. Problem formulation

With the aim of enhancing RUL prediction accuracy using a large amount of unlabeled degradation data, the problem studied is the RUL prediction from the perspective of semi-supervised learning. The training data can be formally expressed as: the labeled training data $D_l = \{(x_i^l, y_i^l) | i = 1, 2, \dots, n_l\}$, and unlabeled training data $D_u = \{x_j^u | j = 1, 2, \dots, n_u\}$, where x_i^l and y_i^l denote the feature variable of the i th training sample, and the corresponding RUL label, respectively, x_j^u denotes the feature variable of the j th unlabeled sample. This paper simultaneously considers labeled data D_l and unlabeled degradation data D_u . In detail, valuable latent information is extracted from unlabeled degradation data to promote the RUL prediction performance. Therefore, the research problem can be formulated as: **under the premise of semi-supervised learning, the RUL prediction is investigated based on an SAM-SVAE.**

3. The methodology of RUL prediction

3.1. Loss design of the sequential variational autoencoder

The fundamental principle of data-driven RUL prediction involves establishing a functional relationship between feature variables x and corresponding RUL values y . As a significant class of data-driven approaches, the core objective of probabilistic generative models is to maximize the conditional probability density $p(y|x)$. For the labeled training data, considering the time dependence of degradation, it is recombined in a sliding time window and then input into the SAM-SVAE. For the convenience of discussion, this section uses a time window data as the object of discussion. On this basis, the probability $p(y_{1:L}|x_{1:L})$ is maximized in a serialized time window, where L is the time window length, which can be expressed as minimizing the loss function in Eq. (1):

$$L = -\log p(y_{1:L} | x_{1:L}) = -\log \int p(y_{1:L}, \phi_{1:L} | x_{1:L}) d\phi_{1:L} \quad (1)$$

where $\phi_{1:L}$ is the latent variable sequence in the probability generation model. To facilitate theoretical derivation, this article introduces an importance distribution $p(\phi_{1:L}|y_{1:L}, x_{1:L})$ to describe the latent variable space. Accordingly, Eq. (1) may be formulated as:

$$\begin{aligned} L &= -\log \int \left[p(y_{1:L}, \phi_{1:L} | x_{1:L}) \frac{p_\theta(\phi_{1:L} | y_{1:L}, x_{1:L})}{p_\theta(\phi_{1:L} | y_{1:L}, x_{1:L})} \right] d\phi_{1:L} \\ &= -\log \int \left[\frac{p(y_{1:L}, \phi_{1:L} | x_{1:L})}{p_\theta(\phi_{1:L} | y_{1:L}, x_{1:L})} p_\theta(\phi_{1:L} | y_{1:L}, x_{1:L}) \right] d\phi_{1:L} \quad (2) \\ &= -\log E_{p_\theta(\phi_{1:L}|y_{1:L}, x_{1:L})} \left[\frac{p(y_{1:L}, \phi_{1:L} | x_{1:L})}{p_\theta(\phi_{1:L} | y_{1:L}, x_{1:L})} \right] \end{aligned}$$

where θ is latent variable space. With Jensen's inequality of convex functions, $f[E(x)] \leq E[f(x)]$ exists, so that Eq. (2) satisfies the following content:

$$L \leq -E_{p_\theta(\phi_{1:L}|y_{1:L}, x_{1:L})} \left[\log \frac{p(y_{1:L}, \phi_{1:L} | x_{1:L})}{p_\theta(\phi_{1:L} | y_{1:L}, x_{1:L})} \right] \quad (3)$$

Let the inequality on the right side be J_s , which can be expanded through Bayesian equation, as shown in the derivation process of

Eq. (4):

$$\begin{aligned}
 J_s &= -\mathbb{E}_{p_\theta(\phi_{1:L}|y_{1:L},x_{1:L})} \left\{ \log \prod_{l=1}^L \left[\frac{p_{\gamma_y}(y_l | y_{1:l-1}, \phi_{1:l-1}, x_{1:L}) p_{\gamma_\phi}(\phi_l | \phi_{1:l-1}, y_{1:l-1}, x_{1:L})}{p_\theta(\phi_l | \phi_{1:l-1}, y_{1:L}, x_{1:L})} \right] \right\} \\
 &= -\mathbb{E}_{p_\theta(\phi_{1:L}|y_{1:L},x_{1:L})} \left\{ \sum_{l=1}^L \log \left[\frac{p_{\gamma_y}(y_l | y_{1:l-1}, \phi_{1:l-1}, x_{1:L}) p_{\gamma_\phi}(\phi_l | \phi_{1:l-1}, y_{1:l-1}, x_{1:L})}{p_\theta(\phi_l | \phi_{1:l-1}, y_{1:L}, x_{1:L})} \right] \right\} \\
 &= -\sum_{l=1}^L \mathbb{E}_{p_\theta(\phi_{1:L}|y_{1:L},x_{1:L})} \left\{ \log \left[\frac{p_{\gamma_y}(y_l | y_{1:l-1}, \phi_{1:l-1}, x_{1:L}) p_{\gamma_\phi}(\phi_l | \phi_{1:l-1}, y_{1:l-1}, x_{1:L})}{p_\theta(\phi_l | \phi_{1:l-1}, y_{1:L}, x_{1:L})} \right] \right\}
 \end{aligned} \quad (4)$$

Considering the nature of the logarithmic function, Eq. (4) can be further split and written in the form described by Eq. (5):

$$\begin{aligned}
 J_s &= -\sum_{l=1}^L \mathbb{E}_{p_\theta(\phi_{1:L}|y_{1:L},x_{1:L})} \left\{ \log \left[p_{\gamma_y}(y_l | y_{1:l-1}, \phi_{1:l-1}, x_{1:L}) \right] \right\} \\
 &\quad - \sum_{l=1}^L \mathbb{E}_{p_\theta(\phi_{1:L}|y_{1:L},x_{1:L})} \left\{ \log \left[\frac{p_{\gamma_\phi}(\phi_l | \phi_{1:l-1}, y_{1:l-1}, x_{1:L})}{p_\theta(\phi_l | \phi_{1:l-1}, y_{1:L}, x_{1:L})} \right] \right\} \\
 &= -\sum_{l=1}^L \mathbb{E}_{p_\theta(\phi_{1:L}|y_{1:L},x_{1:L})} \left\{ \log \left[p_{\gamma_y}(y_l | y_{1:l-1}, \phi_{1:l-1}, x_{1:L}) \right] \right\} \\
 &\quad + \sum_{l=1}^L \mathbb{E}_{p_\theta(\phi_{1:L}|y_{1:L},x_{1:L})} \left\{ \mathbb{E}_{p_\theta(\phi_{1:l-1}|y_{1:l-1},x_{1:L})} \left\{ \log \left[\frac{p_\theta(\phi_l | \phi_{1:l-1}, y_{1:L}, x_{1:L})}{p_{\gamma_\phi}(\phi_l | \phi_{1:l-1}, y_{1:l-1}, x_{1:L})} \right] \right\} \right\}
 \end{aligned} \quad (5)$$

where γ_y and γ_ϕ are the latent space that generates y and ϕ , respectively. Combined with the mathematical properties of KL divergence, Eq. (5) can be written in Eq. (6):

$$\begin{aligned}
 J_s &= -\sum_{l=1}^L \mathbb{E}_{p_\theta(\phi_{1:L}|y_{1:L},x_{1:L})} \left\{ \log \left[p_{\gamma_y}(y_l | y_{1:l-1}, \phi_{1:l-1}, x_{1:L}) \right] \right\} \\
 &\quad + \sum_{l=1}^L \mathbb{E}_{p_\theta(\phi_{1:L}|y_{1:L},x_{1:L})} \left\{ D_{KL} \left[p_\theta(\phi_l | \phi_{1:l-1}, y_{1:L}, x_{1:L}) \parallel \right. \right. \\
 &\quad \times \left. \left. p_{\gamma_\phi}(\phi_l | \phi_{1:l-1}, y_{1:l-1}, x_{1:L}) \right] \right\}
 \end{aligned} \quad (6)$$

Eq. (6) reveals that the overall deep network's loss function J decomposes into a negative logarithmic component and a KL divergence term, which are represented by $J_1^{(s)}$ and $J_2^{(s)}$, respectively:

$$J_1^{(s)} = -\sum_{l=1}^L \mathbb{E}_{p_\theta(\phi_{1:L}|y_{1:L},x_{1:L})} \left\{ \log \left[p_{\gamma_y}(y_l | y_{1:l-1}, \phi_{1:l-1}, x_{1:L}) \right] \right\} \quad (7)$$

$$J_2^{(s)} = \sum_{l=1}^L \mathbb{E}_{p_\theta(\phi_{1:L}|y_{1:L},x_{1:L})} \left\{ D_{KL} \left[p_\theta(\phi_l | \phi_{1:l-1}, y_{1:L}, x_{1:L}) \parallel p_{\gamma_\phi}(\phi_l | \phi_{1:l-1}, y_{1:l-1}, x_{1:L}) \right] \right\} \quad (8)$$

The above derivation process is crucial to guide the construction of deep neural network. Consequently, this paper first designs the loss function of the sequential variational autoencoder. It can be seen that the construction of the probabilistic deep generation network specifically includes three sub-networks, namely, the inference sub-network, the encoding sub-network and the decoding sub-network. To describe the uncertainty in the RUL prediction process, combined with the properties of the probability generation network, the output of each sub-network corresponds to a Gaussian distribution, namely, the mean and variance, as shown in Eqs. (9)–(11):

$$p_\theta(\phi_l | \phi_{1:l-1}, y_{1:L}, x_{1:L}) = N(u_l^{(\phi)}, \sigma_l^{(\phi)}) \quad (9)$$

$$p_{\gamma_\phi}(\phi_l | \phi_{1:l-1}, y_{1:l-1}, x_{1:L}) = N(u_l^{(\hat{\phi})}, \sigma_l^{(\hat{\phi})}) \quad (10)$$

$$p_{\gamma_y}(y_l | y_{1:l-1}, \phi_{1:l-1}, x_{1:L}) = N(u_l^{(y)}, \sigma_l^{(y)}) \quad (11)$$

where $u_l^{(\phi)}$ and $\sigma_l^{(\phi)}$ denote the mean and variance of the inference sub-network, respectively, $u_l^{(\hat{\phi})}$ and $\sigma_l^{(\hat{\phi})}$ denote the mean and variance of the encoding sub-network, respectively, $u_l^{(y)}$ and $\sigma_l^{(y)}$ denote the mean and

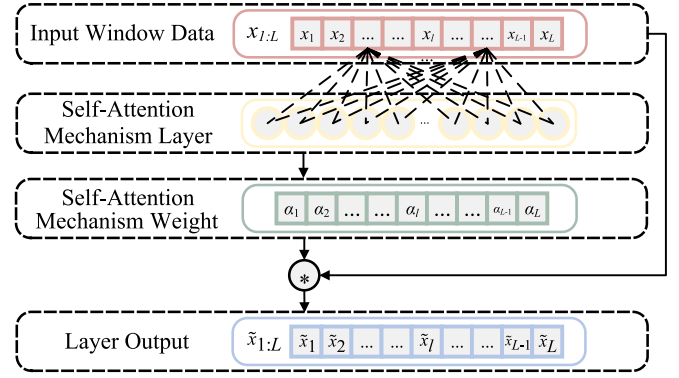


Fig. 1. The framework of SAM layer.

variance of the decoding sub-network, respectively. It is worth mentioning that this paper explores the correspondence between a Bayesian filtering and the state space model in control theory. Particularly, the encoding sub-network represents the state transition equation within the state space model. Similarly, the decoding sub-network embodies the measurement equation in this model. After introducing the loss function design of the SVAE, it is necessary to introduce the specific network design.

3.2. Design of SAM-SVAE network

Regarding the network input feature variables $x_{1:L}$ and the corresponding RUL label $y_{1:L}$, a BiGRU based on the self-attention mechanism (SAM) is designed for encoding. The reason for using the SAM layer is to focus on the key parts of the input time window, which can be shown in Fig. 1.

Regarding the feature encoding forward network, the mapping relationship between different time instances and their significance α_l can be obtained through the SAM layer, as shown in Eq. (12):

$$\alpha_l = \frac{e^{w_l x_l + b_l}}{\sum_{l=1}^L e^{w_l x_l + b_l}} \quad (12)$$

where x_l denotes the input sample at the l th moment, w_l and b_l are the corresponding weight and bias. After obtaining the significance at each moment, the normalized importance is multiplied to acquire the output of the forward SAM layer, the expression is shown in Eq. (13):

$$\tilde{X} = [\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_L] = [\alpha_1 x_1, \alpha_2 x_2, \dots, \alpha_L x_L] \quad (13)$$

The SAM layer's output is fed into the gated recurrent unit (GRU), a recurrent neural network variant equipped with reset and update gates. These two gates help control the flow of temporal information. Corresponding content of GRU can refer to [38]. The output can be obtained through the forward SAM-GRU, as shown in Eq. (14):

$$h_f^{(x)} = GRU(\tilde{X}) \quad (14)$$

To provide more comprehensive temporal information so that the model can better mine the hidden characteristic. This paper designs a backward SAM-GRU. In detail, the input data are encoded in reverse order from the last time to the first time, and then spliced with the output of the forward GRU in Eq. (14). In this sense, the weight of the backward SAM-GRU can be shown in Eq. (15):

$$\beta_j = \frac{e^{w_j [h_f, \tilde{X}]_j + b_j}}{\sum_{j=1}^{d+L} e^{w_j [h_f, \tilde{X}]_j + b_j}} \quad (15)$$

where d denotes the hidden layer dimension of the forward GRU, w_j and b_j denote the corresponding weight and bias of the j th dimension of the splicing matrix, respectively. On this basis, the encoding output

$h_b^{(x)}$ of the SAM-BiGRU can be obtained, as shown in Eq. (16):

$$h_b^{(x)} = GRU [\beta_1 h_1, \beta_2 h_2, \dots, \beta_d h_d, \beta_{d+1} \tilde{x}_1, \beta_{d+2} \tilde{x}_2, \dots, \beta_{d+L} \tilde{x}_L] \quad (16)$$

Similar to the feature variable, the RUL label can be forward-encoded and bidirectional-encoded to obtain $h_f^{(y)}$ and $h_b^{(y)}$. These two ways correspond to the label encoding under the a priori and posterior conditions, respectively. The encoding under the a priori condition only considers the labels before the time instance l , and the labels in the later moments need to be masked. By obtaining the feature encoding matrix and the label encoding matrix, the inference model under the posterior conditions described in Eq. (9) can be constructed. Specifically, this paper constructs a Gaussian distribution network, namely, GussianNet to describe the inference model, which can be shown in Eq. (17):

$$\begin{bmatrix} u_l^{(\phi)} \\ \sigma_l^{(\phi)} \end{bmatrix} = \text{GussianNet}^{(1)} \left(h_b^{(x)}, h_b^{(y)} \right) \quad (17)$$

where GussianNet is implemented based on a multi-layer perceptron, which accomplishes the mapping relationship from the encoding matrix to the mean, and standard deviation of the latent variable through the fully connected layers. The posterior probability $p_\theta(\phi_l | \phi_{1:l-1}, y_{1:L}, x_{1:L})$ can be obtained.

Remark 1. This inference sub-network only uses the time window data and corresponding labels in the training phase, to tighten the lower bound according to theoretical derivation, which does not participate in forward reasoning in the testing phase.

Furthermore, by sampling the Gaussian distribution through $u_l^{(\phi)}$, $\sigma_l^{(\phi)}$, and initializing the latent variables, the latent variable sequence $\phi_{0:l-1}$ can be obtained. Moreover, to achieve the latent variable forward prediction in Eq. (10), similar to the state transition equation pattern in the state space theory, a Gaussian distribution network is designed to construct the encoding sub-network, as shown in Eq. (18):

$$\begin{bmatrix} u_l^{(\phi)} \\ \sigma_l^{(\phi)} \end{bmatrix} = \text{GussianNet}^{(2)} \left(h_b^{(x)}, h_f^{(y)}, \phi_{0:l-1} \right) \quad (18)$$

Remark 2. During supervised training stage, the encoding sub-network may optionally take the historical labels $y_{1:l-1}$ as auxiliary context to stabilize the KL pairing with the inference sub-network, and to regularize the latent transition. During testing stage, the encoding sub-network is label-free. The input related to the label is blocked, ensuring feasible forward propagation without any access to ground-truth labels.

The $p_{\gamma_\phi}(\phi_l | \phi_{1:l-1}, y_{1:l-1}, x_{1:L})$ can be obtained by Eq. (18) in the paper. Combined with the probability function $p_\theta(\phi_l | \phi_{1:l-1}, y_{1:L}, x_{1:L})$ in Eq. (17), the KL divergence can be calculated in Eq. (8). According to the mean and standard deviation in Eq. (17), Gaussian distribution sampling is performed. The decoding sub-network is designed, as shown in Eq. (19):

$$\begin{bmatrix} u_l^{(y)} \\ \sigma_l^{(y)} \end{bmatrix} = \text{GussianNet}^{(3)} \left(h_b^{(x)}, h_f^{(y)}, \phi_{1:l} \right) \quad (19)$$

Remark 3. During supervised training, the decoding sub-network is conditioned on $y_{1:l-1}$ for teacher forcing and conditional reconstruction to improve stability. During testing stage, the decoding sub-network is label-free. The input related to the label is blocked. The prediction is produced from the latent state and the online time window data.

During the training process, the encoder and decoder sub-networks are constrained to have the available information set at time l as $I_l = \{x_{1:l-1}, y_{1:l-1}, \phi_{1:l-1}\}$. This means that when generating the distribution over $\hat{\phi}_l$ and y_l , the encoder and decoder sub-networks only have access to the label, observed inputs, and the previous latent states before l . Any labels beyond $t \leq l$ are considered future information and

are unavailable. Intuitively, this mask renders labels at times not yet known invisible to the network, eliminating information leakage at the source. The inference sub-network is only visible to the full-window labels and inputs during the training phase to tighten the lower bound of the evidence. It does not participate in forward reasoning during the testing and deployment phases, so that the whole online prediction process is causal.

According to the mean and variance in Eq. (19), the probability $p_{\gamma_y}(y_l | y_{1:l-1}, \phi_{1:l-1}, x_{1:L})$ can be acquired. Through Eq. (7), the corresponding negative logarithm term in loss function is calculated. The corresponding results are combined with Eq. (8) to obtain the loss function of the whole SAM-SVAE, which is used for parameter update in the training process. The framework of the SAM-SVAE is shown in Fig. 2. It should be emphasized that since the decoder sub-network produces the RUL's mean and variance, the uncertainty in degradation can be quantified.

3.3. A semi-supervised SAM-SVAE network

Conventional data-driven approaches demand massive labeled samples for supervised model training, which has notable limitations in making full use of unlabeled data. Consequently, this paper conducts a research on RUL prediction from a semi-supervised perspective by simultaneously utilizing labeled and unlabeled data samples. Regarding the labeled training data $D_l = \{(x_i^l, y_i^l) | i = 1, 2, \dots, n_l\}$, and the unlabeled training data $D_u = \{x_j^u | u = 1, 2, \dots, n_u\}$. The labeled training data carries out the training according to Section 3.2.

On the other hand, the unsupervised SAM-SVAE and the supervised SAM-SVAE generally have the same network form. Different from the modeling form in supervised learning, the unsupervised models pay more attention to the data generation mechanism. In this sense, the data generation process is simulated. On this basis, for unlabeled training data, the inference sub-network, encoding sub-network, and decoding sub-network in Eqs. (9)–(11) need to be reorganized, as shown in Eqs. (20)–(22):

$$p_\theta(\tilde{\phi}_l | \tilde{\phi}_{1:l-1}, \tilde{x}_{1:L}) = N(u_l^{(\tilde{\phi})}, \sigma_l^{(\tilde{\phi})}) \quad (20)$$

$$p_{\gamma_{\tilde{\phi}}}(\tilde{\phi}_l | \tilde{\phi}_{1:l-1}, \tilde{x}_{1:l-1}) = N(u_l^{(\tilde{\phi})}, \sigma_l^{(\tilde{\phi})}) \quad (21)$$

$$p_{\gamma_x}(\tilde{x}_l | \tilde{\phi}_{1:l-1}, \tilde{x}_{1:l-1}) = N(u_l^{(x)}, \sigma_l^{(x)}) \quad (22)$$

where $u_l^{(\tilde{\phi})}$ and $\sigma_l^{(\tilde{\phi})}$ denote the mean and variance of the inference sub-network, respectively, $u_l^{(\phi)}$ and $\sigma_l^{(\phi)}$ denote the mean and variance of the encoding sub-network, respectively, $u_l^{(x)}$ and $\sigma_l^{(x)}$ denote the mean and variance of the decoding sub-network, respectively. To be different from the parameter representation under supervised conditions, the characteristic variables and latent variables under unsupervised conditions are uniformly represented by $\tilde{x}$ and $\tilde{\phi}$. It should be noted that the principles of the inference sub-network and the encoding sub-network are consistent with the supervised learning network, except that the RUL label is no longer used as the encoding input of the network, which is replaced by the feature variable. Alternatively, the output of the decoding sub-network under unsupervised conditions is also replaced by the feature variable. Corresponding GussianNets under unsupervised condition are defined as $\text{GussianNet}^{(1)}$, $\text{GussianNet}^{(2)}$, $\text{GussianNet}^{(3)}$. In fact, the SAM-SVAE under unsupervised condition is essentially an autoencoder that learns useful feature representations from input degradation data. Through setting the decoder output to the feature variable, the model learns how to capture key features of the data during the encoding–decoding process, thereby promoting the efficiency of feature extraction. The loss function of the unsupervised

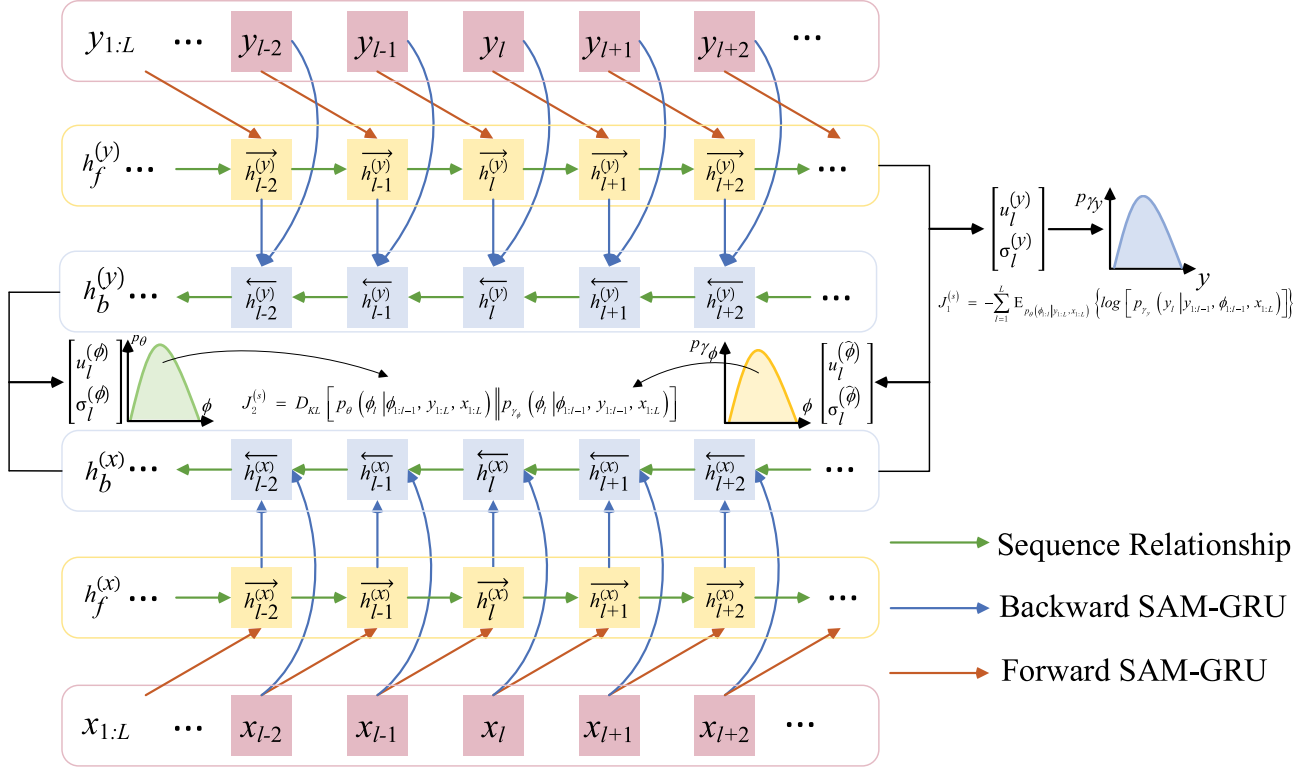


Fig. 2. The framework of the SAM-SVAE.

SAM-SVAE is shown in Eq. (23):

$$J_u = - \sum_{l=1}^L E_{p_{\theta}(\tilde{\phi}_{1:l} | \tilde{x}_{1:L})} \{ \log [p_{\gamma_x}(\tilde{x}_l | \tilde{\phi}_{1:l-1}, \tilde{x}_{1:l-1})] \} \\ + \sum_{l=1}^L E_{p_{\theta}(\tilde{\phi}_{1:l-1} | \tilde{x}_{1:L})} \{ D_{KL} [p_{\theta}(\tilde{\phi}_l | \tilde{\phi}_{1:l-1}, \tilde{x}_{1:L}) \| p_{\gamma_{\phi}}(\tilde{\phi}_l | \tilde{\phi}_{1:l-1}, \tilde{x}_{1:l-1})] \} \quad (23)$$

The framework of the SAM-SVAE from the unsupervised perspective is shown in Fig. 3 in the paper. The proposed approach does not generate synthetic labels. The unsupervised branch does not perform pseudo-labeling. The contribution of the unsupervised path is to learn representations, and regularize the model through a reconstruction and KL term, which are trained jointly with the supervised likelihood.

Consequently, combined with the loss function of SAM-SVAE under the supervised condition in Eq. (8), the loss function of the SAM-SVAE based on the semi-supervised perspective can be shown in Eq. (24).

$$J = J_s + \lambda J_u \\ = - \sum_{l=1}^L E_{p_{\theta}(\phi_{1:l} | y_{1:L}, x_{1:L})} \{ \log [p_{\gamma_y}(y_l | y_{1:l-1}, \phi_{1:l-1}, x_{1:L})] \} \\ + \sum_{l=1}^L E_{p_{\theta}(\phi_{1:l-1} | y_{1:L}, x_{1:L})} \{ D_{KL} [p_{\theta}(\phi_l | \phi_{1:l-1}, y_{1:L}, x_{1:L}) \| \\ \times p_{\gamma_{\phi}}(\phi_l | \phi_{1:l-1}, y_{1:l-1}, x_{1:L})] \} \\ - \lambda \sum_{l=1}^L E_{p_{\theta}(\tilde{\phi}_{1:l} | \tilde{x}_{1:L})} \{ \log [p_{\gamma_x}(\tilde{x}_l | \tilde{\phi}_{1:l-1}, \tilde{x}_{1:l-1})] \} \\ + \lambda \sum_{l=1}^L E_{p_{\theta}(\tilde{\phi}_{1:l-1} | \tilde{x}_{1:L})} \{ D_{KL} [p_{\theta}(\tilde{\phi}_l | \tilde{\phi}_{1:l-1}, \tilde{x}_{1:L}) \| p_{\gamma_{\phi}}(\tilde{\phi}_l | \tilde{\phi}_{1:l-1}, \tilde{x}_{1:l-1})] \} \quad (24)$$

Combining Eqs. (9)–(11) and Eqs. (20)–(22), it is not difficult to see that the distribution described by Eq. (10) in the supervised encoding

sub-network is consistent with Eq. (20) in the supervised encoding sub-network, which means that the unsupervised model and the supervised model adopt the same structural for modeling the conditional distribution. In this sense, the network parameters can be shared between these two tasks. Consequently, a large amount of unlabeled degraded data may become useful, which promotes the model generalization performance and efficiency. The framework of the SAM-SVAE from the semi-supervised perspective is shown in Fig. 4. Algorithm 1 describes supervised SAM-SVAE training algorithm. Algorithm 2 describes semi-supervised SAM-SVAE training algorithm (supervised branch omitted).

4. Case study

4.1. Dataset introduction and research task

The research utilizes NASA's aircraft turbofan engine dataset to evaluate the SAM-SVAE's RUL prediction performance from a semi-supervised viewpoint. The dataset comprises engine operation data under multiple working conditions, which includes multi-source sensor measurement data, such as temperature, pressure, vibration, etc., and the performance parameters, a total of 26 variables. Under every operational scenario, the dataset is partitioned into training and testing subsets. The training set contains life cycle data of multiple engines, which is used for model training and parameter adjustment. The testing set includes incomplete degradation sequence of multiple engines. The model performance is evaluated by predicting the RUL at the last time point of the sequence. The complete dataset configuration is presented in Table 1. The detailed introduction to the dataset can be referred to [39].

Regarding the research task, this investigation focuses on examining the RUL prediction capability of SAM-SVAE within a semi-supervised

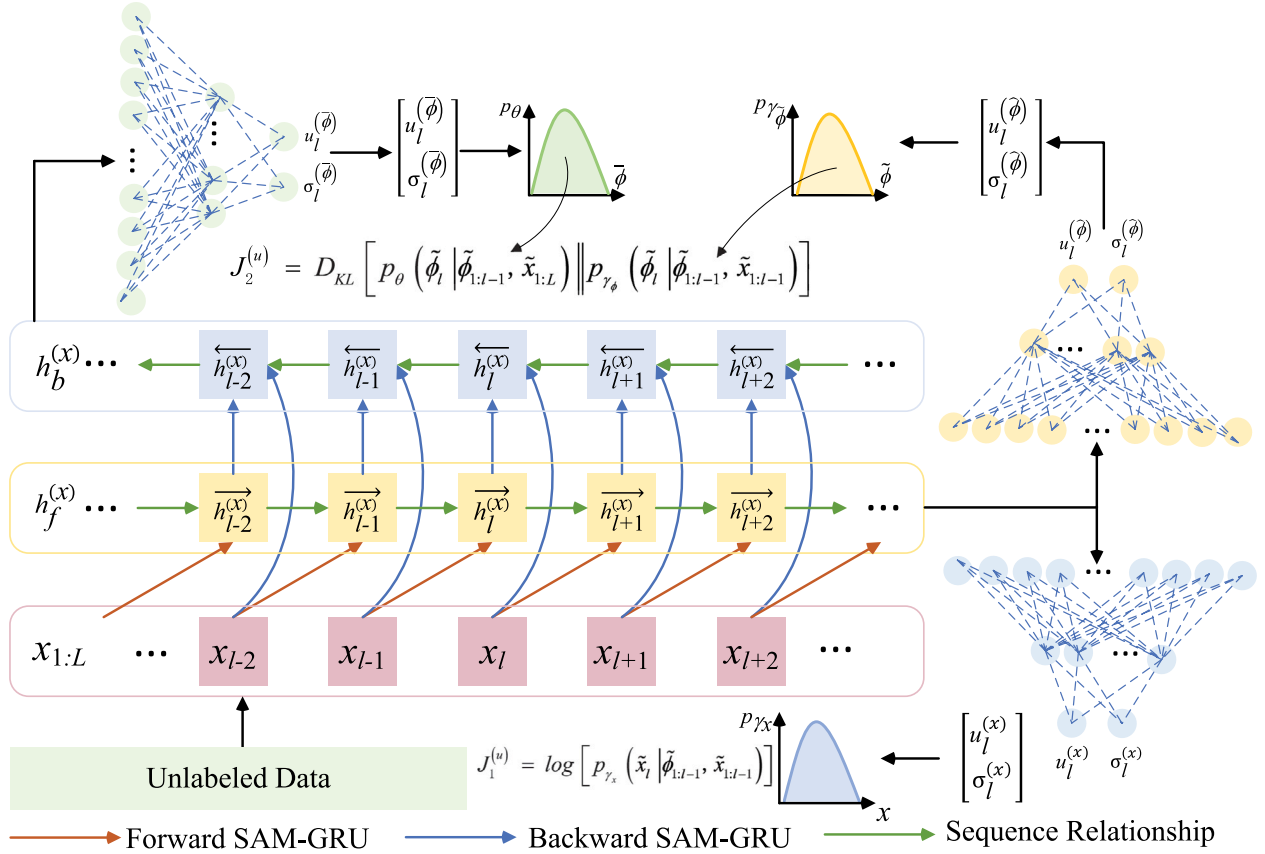


Fig. 3. The framework of the SAM-SVAE from the unsupervised perspective.

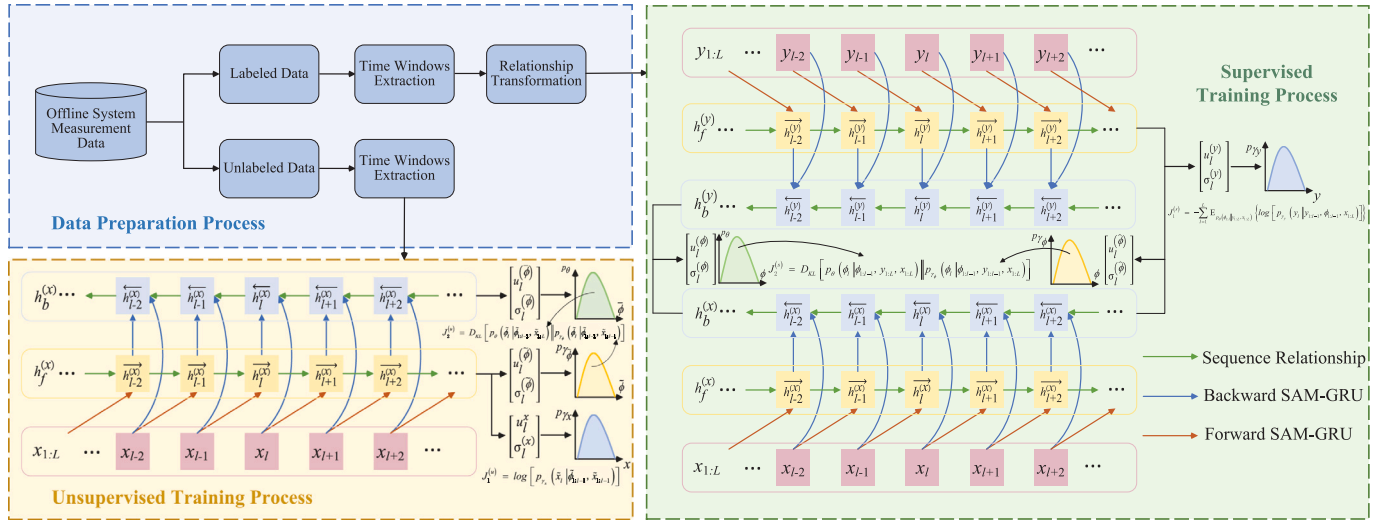


Fig. 4. The framework of the SAM-SVAE from the semi-supervised perspective.

learning framework. When training the semi-supervised model, this paper retain the training data at different percentages (2%, 5%, 10%, 50% and 100%) for each operational condition to evaluate the network performance. It should be noted that different percentages of training data are randomly selected. Furthermore, 80% of the training data are randomly selected in each sub-dataset for training, and the remaining 20% of the data are used for verification. In this sense,

the training sample counts at selected label percentages are shown in Table 2. No synthetic data or pseudo-labels are generated at any stage. The semi-supervised branch relies solely on unlabeled time window data extracted from the original sequences. The contribution of the unsupervised path is to learn representations, and regularize the model through a reconstruction and KL term, which are trained jointly with the supervised likelihood.

Algorithm 1 Supervised SAM-SVAE training algorithm

REQUIRE: Labeled data $D_l = \{(x_i^l, y_i^l) | i = 1, 2, \dots, n_l\}$, window length L , fixed seeds.

ENSURE: Learned parameters of GaussianNet(1/2/3) (supervised), and the SAM-BiGRU encoder.

- 1: **for** each minibatch $(x_{1:L}, y_{1:L})$ sampled from D_l **do**
- 2: Forward SAM on features: compute attention weights over $x_{1:L}$ and reweight the feature window. (Eqs. (12)-(13))
- 3: Forward SAM-GRU on features: feed the reweighted sequence to a GRU to obtain $h_f^{(x)}$. (Eq. (14))
- 4: Backward SAM weights: form the concatenation of $h_f^{(x)}$ and the weighted features, compute backward weighting coefficients. (Eq. (15))
- 5: BiGRU feature encoding: apply a GRU to the weighted concatenation to obtain $h_b^{(x)}$. (Eq. (16))
- 6: Label encodings: analogously process $y_{1:L}$ to obtain $h_f^{(y)}$ (forward) and $h_b^{(y)}$ (backward).
- 7: **for** $l = 1$ to L **do**
- 8: Inference sub-network: use GaussianNet⁽¹⁾ with inputs $(h_b^{(x)}, h_b^{(y)})$ to produce the posterior parameters for ϕ_l , sample ϕ_l via reparameterization. (Eq. (17))
- 9: Encoding sub-network: use GaussianNet⁽²⁾ with inputs $(h_b^{(x)}, h_f^{(y)}, \phi_{0:l-1})$ to produce the prior parameters for ϕ_l , ensure $h_f^{(y)}$ masks labels $y_{l \geq l}$. (Eq. (18))
- 10: Decoding sub-network: use GaussianNet⁽³⁾ with inputs $(h_b^{(x)}, h_f^{(y)}, \phi_{1:l})$ to produce the predictive distribution for y_l , ensure $h_f^{(y)}$ masks labels $y_{l \geq l}$. (Eq. (19))
- 11: **end for**
- 12: Supervised loss: compute J_s as the sum over $l = 1, \dots, L$ of the negative log-likelihood term for y_l (likelihood of GaussianNet⁽³⁾), plus the KL divergence between the posterior (GaussianNet⁽¹⁾) and the prior (GaussianNet⁽²⁾). (Eqs. (7)–(8))
- 13: Update parameters of GaussianNet(1/2/3), and the SAM-BiGRU encoder by minimizing J_s .
- 14: **end for**

Table 1

The dataset configuration of NASA dataset.

Dataset	Engine datasets			
	FD001	FD002	FD003	FD004
Training engines	100	260	100	249
Testing engines	100	259	100	248
Operational modes	1	6	1	6
Failure modes	1	1	2	2

Table 2

Training sample counts at selected label percentages (fill in counts).

Percentage	FD001	FD002	FD003	FD004
2%	220	307	285	392
5%	550	1537	713	1962
10%	1099	3074	1426	3924
50%	5493	15368	7128	19619
100%	10985	30735	14256	39238

4.2. Implementation details and parameter analysis

Considering that sensor data usually have different measurement ranges and units, this paper adopts a data normalization approach based on operational conditions, to normalize the sensor data to the same scale according to the operational conditions. Furthermore, considering that some sensors have constant values during the whole operational process, this paper filters the sensor data, and eliminates

Algorithm 2 Semi-supervised SAM-SVAE training algorithm (supervised branch omitted)

REQUIRE: Labeled data D_l , unlabeled data $D_u = \{x_i^u | i = 1, \dots, n_u\}$, window length L , fixed seeds.

ENSURE: Learned parameters of GaussianNet(1/2/3), GaussianNet($\tilde{1}/\tilde{2}/\tilde{3}$) (supervised & unsupervised), and the SAM-BiGRU encoders (supervised & unsupervised).

- 1: **for** each minibatch $\tilde{x}_{1:L}$ sampled from D_u **do**
- 2: Forward SAM on $\tilde{x}_{1:L}$ to obtain attention weights and reweight the window. (Eqs. (12)-(13))
- 3: Forward SAM-GRU on the reweighted sequence to obtain $h_f^{(\tilde{x})}$. (Eq. (14))
- 4: Backward SAM weights on the concatenation of $h_f^{(\tilde{x})}$ and the reweighted features. (Eq. (15))
- 5: BiGRU feature encoding to obtain $h_b^{(\tilde{x})}$. (No label encodings are computed in this branch.) (Eq. (16))
- 6: **for** $l = 1$ to L **do**
- 7: Unsupervised inference sub-network: use GaussianNet⁽¹⁾ with current feature encodings, and past latent states to produce posterior parameters for $\tilde{\phi}_l$, sample $\tilde{\phi}_l$ via reparameterization. (Eq. (20))
- 8: Unsupervised encoding sub-network: use GaussianNet⁽²⁾ with $(h_b^{(\tilde{x})}, \tilde{\phi}_{1:l-1})$ to produce prior parameters for $\tilde{\phi}_l$. (Eq. (21))
- 9: Unsupervised decoding sub-network: use GaussianNet⁽³⁾ to generate the distribution for $\tilde{x}_l$ from $(\tilde{\phi}_{1:l-1}, \tilde{x}_{1:l-1})$ for reconstruction. (Eq. (22))
- 10: **end for**
- 11: Unsupervised loss: compute J_u as the sum over $l = 1, \dots, L$ of the reconstruction loss for $\tilde{x}_l$ (GaussianNet⁽³⁾), plus the KL divergence between the unlabeled posterior (GaussianNet⁽¹⁾) and unlabeled prior (GaussianNet⁽²⁾). (Eq. (23))
- 12: Update parameters of GaussianNet(1/2/3), GaussianNet($\tilde{1}/\tilde{2}/\tilde{3}$), and the SAM-BiGRU encoders by minimizing $J_s + \lambda J_u$. (Eq. (24))
- 13: **end for**

seven types of sensor data with constant values numbered 1, 5, 6, 10, 16, 18, and 19, which promotes the training efficiency of deep neural network. Moreover, this paper adopts a sequence-to-sequence approach. In detail, the time window is designed to combine feature variables and labels to better mine the time dependencies implicit in degradation data.

The final model architecture includes SAM-BiGRU encoder, inference sub-network, encoding sub-network, and decoding sub-network. It should be noted that the weighted trade-off coefficient λ between the supervised task loss and the unsupervised task loss in Eq. (24), the time window length L , the number of BiGRU neurons, the learning rate, the weight decay rate, and the batch size may have an impact on the proposed SAM-SVAE. This section changes the weight coefficient, the time window length, the number of BiGRU neurons, the learning rate, the weight decay coefficient, and the batch size to study the robustness of these hyperparameters for the network. For each C-MAPSS sub-dataset (FD001–FD004), an independent grid search is adopted on a held-out validation split to identify the configuration that optimizes validation performance. 80% of the training data are randomly selected in each sub-dataset for training, and the remaining 20% of the data are used for verification. This per-dataset procedure is motivated by the differences in operating and failure modes across FD001–FD004. The search spaces for all tuned factors are reported in Table 3 in the paper. The resulting optimal configurations for each sub-dataset are summarized in Table 4 in the paper.

The setting of the weighted trade-off coefficient λ reflects the balance between supervised and unsupervised loss in the proposed approach. Specifically, when λ is large, the training process is dominated

Table 3
The parameter adjustment of the grid search.

Parameter	Range	Type
Weighted coefficient	[0.01, 0.1, 1, 2, 5]	Discrete
Time window length	[30, 40, 50, 60, 70, 80, 90]	Discrete
Neuron number	[8, 16, 32, 64, 128]	Discrete
Learning rate	[0.0005, 0.001, 0.005, 0.01, 0.02]	Discrete
Weight decay	[1e-9, 1e-7, 1e-5, 1e-3, 1e-1]	Discrete
Batch size	[32, 64, 128, 256, 512]	Discrete

Table 4
The network configuration of SAM-SVAE.

Parameter	FD001	FD002	FD003	FD004
Weighted coefficient	1	1	1	1
Time window length	70	60	70	50
Neuron number	128	128	128	128
Learning rate	10^{-3}	10^{-3}	0.005	10^{-3}
Weight decay	10^{-7}	10^{-5}	10^{-7}	10^{-5}
Batch size	512	256	512	256
Optimizer	Adam	Adam	Adam	Adam

by the unsupervised reconstruction loss. Although this promotes effective learning of the overall degradation structure, it reduces the model's discriminative ability for the RUL prediction task, leading feature extraction away from the main objective. Conversely, when λ is too small, the model relies excessively on the limited labeled data for parameter optimization, neglecting the contribution of abundant unlabeled degradation data to the latent space representation, which in turn weakens the generalization ability of the semi-supervised framework. Experimental results demonstrate that setting λ to 1 achieves an optimal balance, ensuring both task-oriented feature extraction and effective utilization of structural information from unlabeled data, resulting in the best overall performance across all evaluation metrics.

The length of the temporal window is also a key parameter affecting model performance. If the window is too short, the input contains insufficient information to capture the full dynamics of equipment degradation, resulting in inadequate temporal feature modeling. Conversely, an excessively long window introduces irrelevant or noisy information, increasing the learning burden and risk of overfitting. Experiments show that the window length of 70, 60, 70, 50 for each sub-dataset achieves a good balance between sequence completeness and model complexity.

The number of GRU units in the latent representation is equally important. Too few units limit the model's ability to capture long-term dependencies in sequential data, while too many increase parameter redundancy and the risk of overfitting. Setting the number of GRU units to 128 in this study enables effective extraction of long-term dependencies while maintaining generalization and computational efficiency.

Finally, the learning rate and weight decay are tuned to jointly ensure convergence and regularization. The learning rate of 10^{-3} , 10^{-3} , 0.005, 10^{-3} for each sub-dataset provides efficient optimization. The weight decay coefficient of 10^{-7} , 10^{-5} , 10^{-7} , 10^{-5} for each sub-dataset helps mitigate overfitting and improve generalization. All hyperparameter choices are supported by extensive experiments and sensitivity analysis, ensuring robust and stable prediction performance.

4.3. Ablation experiment of proposed approach

To further validate the effectiveness of each key component in the proposed approach, four groups of ablation experiments are designed. The attention mechanism, the type of recurrent neural network, and the unsupervised branch in the loss function from the network are successively removed, in order to quantitatively assess the contribution of each module to the overall model performance. In detail, to systematically evaluate the contribution of each key component in the proposed approach, four ablation variants are constructed as follows:

M1: Replacing GRU with RNN. The GRU is replaced in the main network with standard RNN cells, enabling assessment of the effect of the recurrent unit design on sequence modeling and prediction performance.

M2: Removing the attention mechanism. The self-attention mechanism is removed, allowing examination of the importance of attention for temporal feature extraction.

M3: Eliminating the unsupervised loss from the objective function. The value of leveraging unlabeled data in the semi-supervised framework is tested.

M4: Replacing the bidirectional GRU with a unidirectional GRU. The role of bidirectional modeling is clarified in capturing degradation dynamics.

The RUL prediction results are evaluated using root mean square error (RMSE) and S-Score, which clearly demonstrate the superiority of the proposed approach. Table 5 presents the ablation results of RMSE and S-Score across approaches, which quantifies the contribution of each core module to the overall performance. As shown, removing the GRU structure (M1) or replacing it with less effective units leads to a marked degradation in both RMSE and S-Score, indicating the importance of effective sequence modeling. The absence of the attention mechanism (M2) or unsupervised loss branch (M3) also results in noticeable performance drops, particularly under more complex operational conditions, demonstrating the value of both temporal feature extraction and semi-supervised learning. Furthermore, discarding the bidirectional structure in GRU (M4) yields moderate yet consistent declines in all metrics. Overall, the proposed SAM-SVAE model consistently achieves the best results across all datasets, underscoring the necessity and effectiveness of each architectural component.

Moreover, to assess the quality of predictive uncertainty under the deployment setting, uncertainty is evaluated with two complementary metrics reported on FD001–FD004. Negative log-likelihood (NLL) is a strictly proper scoring rule for probabilistic prediction that measures the average negative log-probability assigned to held-out observations. Lower NLL indicates a predictive distribution that is better calibrated and appropriately concentrated. Confidence interval coverage (CIC) describes the proportion of observations that fall within the model's central prediction interval at a stated nominal level (e.g., 95%). For 95% CIC, the values nearer the nominal 0.95 indicate superior calibration of the model's predictive uncertainty. Table 6 presents the ablation results of NLL and CIC (95%) across approaches.

Across all four subsets, the proposed SAM-SVAE achieves the lowest NLL and the CIC closest to the nominal 95% among the compared variants, indicating the best balance between calibration and sharpness. On FD001–FD004, CIC for the proposed model is 0.91/0.86/0.87/0.85, whose absolute deviations from 0.95 (4/9/8/10 percentage points) are smaller than any ablation. This aligns with the RMSE and S-Score gains. It is shown that attention-based dynamic serialization, bidirectional encoding, and the semi-supervised objective jointly improve distributional quality, not only point accuracy. The uncertainty study confirms that the full SAM-SVAE improves probabilistic quality in a way that mirrors its point-prediction advantages.

4.4. Experimental result analysis

Facing supervised condition, for FD001–FD004 of the aircraft turbofan engine dataset, this paper constructs the SAM-SVAE model for algorithm verification. The training stage utilizes the designated hyperparameters (Table 4) to fit the model to the training dataset. The model is further validated through corresponding test data of turbofan engines.

To verify the advantages of the proposed approach, multiple causal baselines are designed for fair comparison, including temporal convolutional network with causal mask (TCNCM), gated recurrent unit with causal mask (GRUCM), and Transformer with causal attention (Transformer-CA). All the approaches share identical preprocessing,

Table 5

The ablation results of RMSE and S-Score across approaches.

Approach	RMSE				S-score			
	FD001	FD002	FD003	FD004	FD001	FD002	FD003	FD004
M1	15.33	19.94	15.45	20.51	313.62	2243.17	310.24	3045.83
M2	14.42	18.21	14.28	18.67	258.39	1765.74	269.89	2385.16
M3	14.91	18.92	14.97	19.59	281.51	1626.93	284.92	2322.61
M4	13.97	17.56	13.66	17.82	241.76	1324.88	263.48	1714.44
Proposed SAM-SVAE	12.38	15.91	12.53	16.48	208.63	921.29	225.69	1636.99

Table 6

The ablation results of NLL and CIC (95%) across approaches.

Approach	NLL				CIC			
	FD001	FD002	FD003	FD004	FD001	FD002	FD003	FD004
M1	4.55	5.22	5.05	7.78	0.86	0.80	0.83	0.78
M2	4.18	4.74	4.79	7.18	0.88	0.82	0.85	0.81
M3	4.27	4.88	4.89	7.32	0.88	0.80	0.84	0.82
M4	4.02	4.51	4.58	6.92	0.90	0.83	0.86	0.84
Proposed SAM-SVAE	3.89	4.32	4.43	6.74	0.91	0.86	0.87	0.85

Table 7

The compared results of RMSE and S-Score for typical causal approaches.

Approach	RMSE				S-score			
	FD001	FD002	FD003	FD004	FD001	FD002	FD003	FD004
TCNCM	14.51	19.05	15.07	21.90	302.72	1548.00	368.98	2469.00
GRUCM	14.32	19.35	14.96	21.20	285.89	1634.00	336.17	2399.50
Transformer-CA	13.96	16.88	14.25	18.45	250.91	1373.00	275.44	2009.20
Proposed SAM-SVAE	12.38	15.91	12.53	16.48	208.63	921.29	225.69	1636.99

sliding-window construction, splits, and fixed seeds. The causal masks and attention prohibit any access to future inputs or labels during training and inference. Table 7 presents the compared results of RMSE and S-Score for typical causal approaches. Among these causal baselines, Transformer-CA is strongest, yet SAM-SVAE attains consistent gains across all subsets. Relative to Transformer-CA, RMSE decreases by approximately 12.7%, 6.1%, 13.7%, and 12.0% for FD001–FD004, respectively. TCNCM and GRUCM trail further, particularly on FD002 and FD004 where longer and more heterogeneous operational cycles increase the burden on temporal credit assignment. Purely convolutional and uni-gate recurrent baselines are less effective than causal attention, which is still remain behind SAM-SVAE.

Considering that the operational conditions of FD002 and FD004 are complicated, this paper represents the RUL prediction results of the proposed SAM-SVAE in FD002 and FD004 in Fig. 5 and Fig. 6, respectively. At the same time, to demonstrate the benefits of SAM-SVAE, the RUL prediction approach is compared with several recently published approaches. The experimental comparison results of RMSE and S-Score are presented in Table 8. As demonstrated in Table 8 and Figs. 5–6, the proposed SAM-SVAE achieves outstanding prediction performance.

Facing semi-supervised condition, this paper randomly retain training data at different percentages (2%, 5%, 10%, 30%, 50% and 100%) for each operational condition to evaluate the network performance under different data availability. The summary of experimental results under the whole semi-supervised condition is shown in Table 9. By comparing the model performance under different supervised percentages, the experimental results demonstrate that with merely 2% labeled data, the model achieves the poorest RUL prediction accuracy on the test set. To further elucidate the impact of label proportion on model performance, a systematic analysis is conducted based on the learning mechanism and the characteristics of the semi-supervised framework. When the proportion of labeled data is low (e.g., 2% or 5%), the model relies primarily on the reconstruction error from unlabeled samples, as supervised signals are extremely limited. In this scenario, the unsupervised branch helps capture temporal dependencies and distributional

structure in the degradation data, but it is less effective at learning the complex mapping between features and remaining useful life, leading to constrained prediction accuracy. As the proportion of labeled data increases, more direct supervision from true RUL labels enhances parameter optimization and significantly improves both generalization and prediction accuracy, as evidenced by the consistent improvements in RMSE and S-Score. Notably, when the label proportion exceeds 30%, model performance under various operating conditions approaches that of fully supervised training, demonstrating high label efficiency of the proposed semi-supervised framework. On the other hand, even with extremely low labeling rates (such as 2%), the model still achieves reasonable results in Table 9, owing to the unsupervised branch in the sequential variational autoencoder, which fully leverages abundant unlabeled degradation data to extract and represent temporal degradation features. The encoding–decoding mechanism in the latent space allows limited labeled samples to provide effective calibration for RUL prediction.

To further assess the quality of the predictive uncertainty beyond scalar metrics such as NLL and CIC, the calibration curves and probability integral transform (PIT) histograms are plotted for all four operational conditions. For each sub-dataset, the symmetric predictive intervals at multiple nominal coverage levels (for example from 0.4 up to 0.95) are constructed from the predictive distribution of the proposed SAM-SVAE. The corresponding empirical coverage is computed as the proportion of test observations, whose true RUL falls inside these intervals. The calibration curves of FD001–FD004 are shown in Fig. 7 in the paper. The calibration curve then links the nominal coverage on the horizontal axis with the empirical coverage on the vertical axis, while the diagonal line represents perfect calibration. The resulting calibration curves for FD001–FD004 stay close to the diagonal reference line over a wide range of nominal levels. This indicates that the predictive intervals produced by SAM-SVAE neither systematically under-cover nor over-cover the true RUL.

In addition, PIT histograms are drawn by mapping each test observation to its cumulative probability under the corresponding predictive

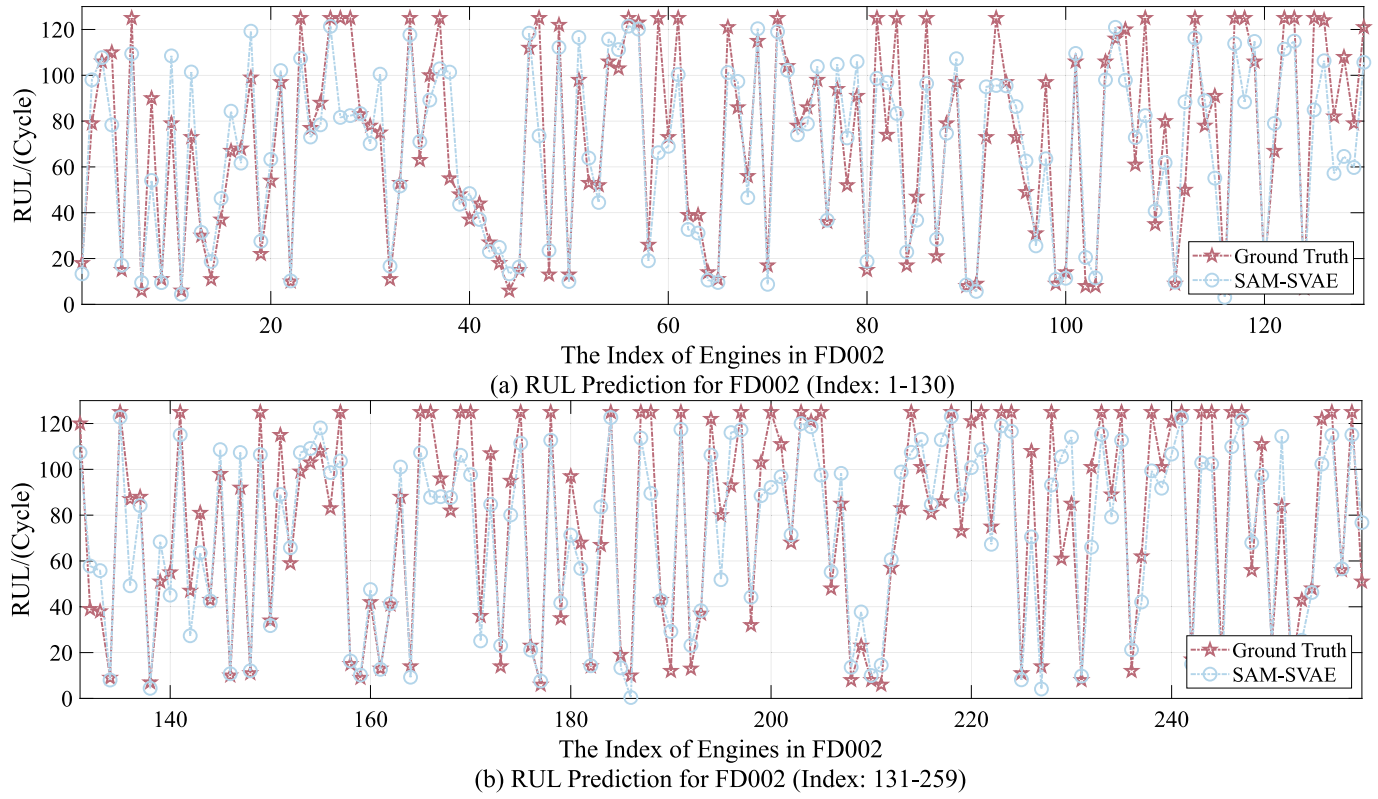


Fig. 5. Experimental results of the proposed SAM-SVAE in FD002.

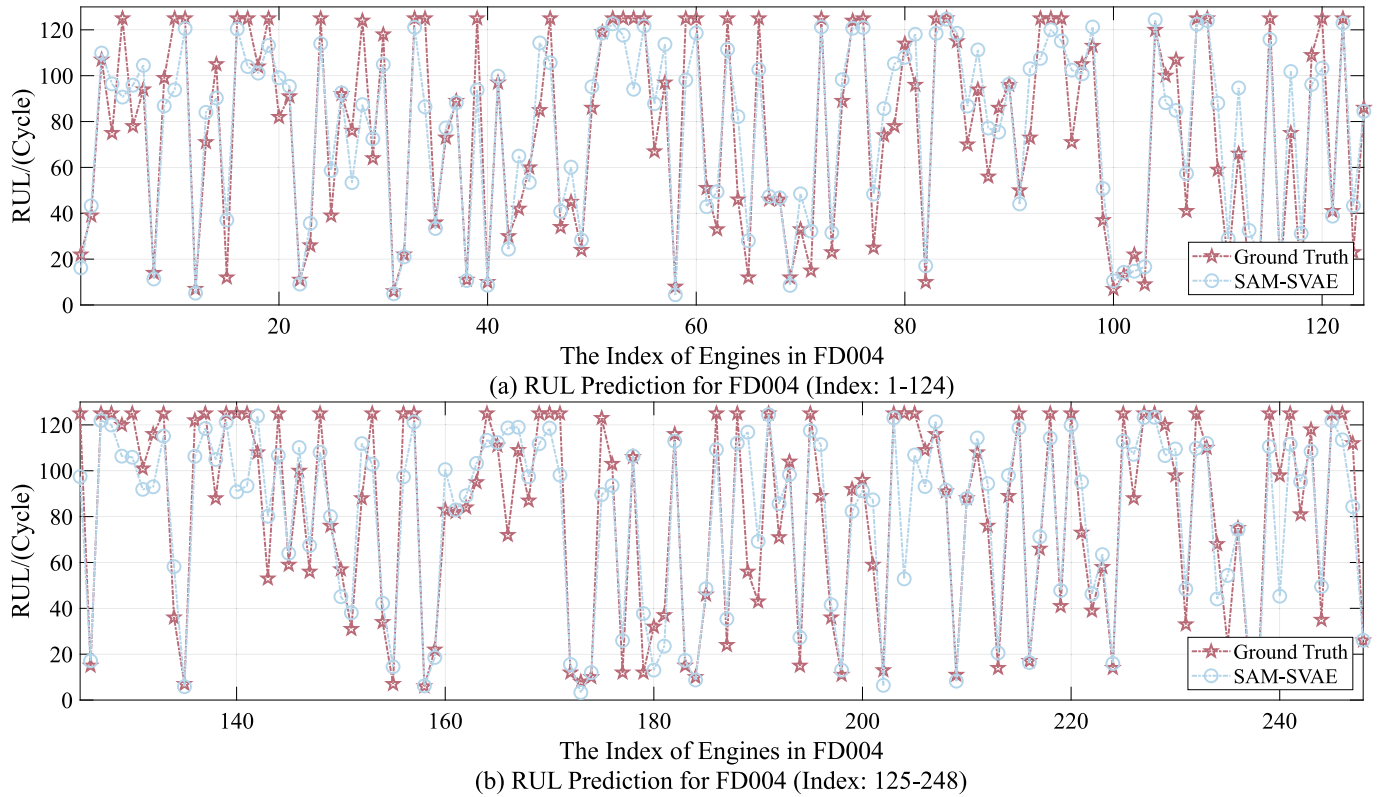


Fig. 6. Experimental results of the proposed SAM-SVAE in FD004.

Table 8

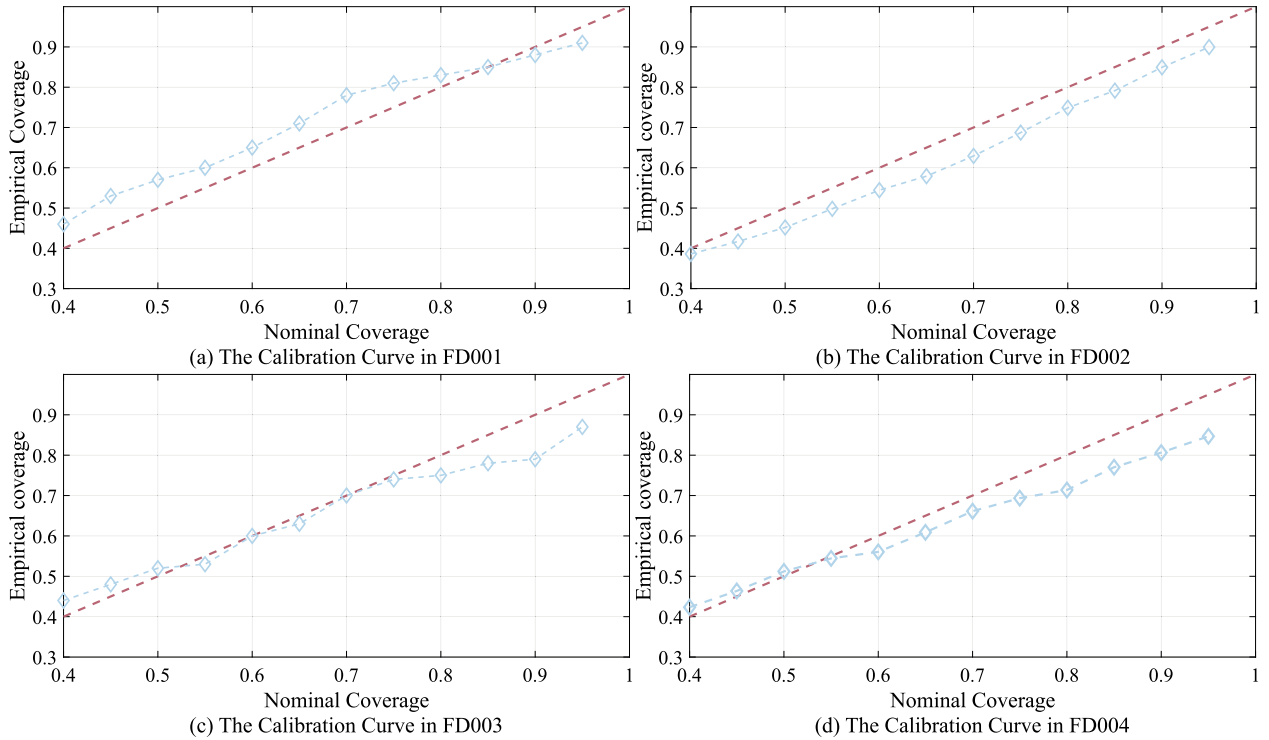
The evaluation of RUL prediction performance across approaches.

Approach	RMSE				S-Score			
	FD001	FD002	FD003	FD004	FD001	FD002	FD003	FD004
VAE-RNN [40]	15.81	24.12	14.88	26.54	326.00	4183.00	722.00	5634.00
CEED-DLSTM [41]	14.72	29.00	17.72	33.43	262.00	6953.00	452.00	15069.00
IESGP [42]	14.72	24.81	14.99	28.61	331.90	4245.40	355.20	6280.80
Attention-LSTM [43]	14.53	–	–	27.06	322.44	–	–	5649.14
CNN-LSTM-DA [44]	14.40	27.23	14.32	26.69	290.00	9869.00	316.00	6594.00
BiGRU [45]	–	26.54	–	29.13	–	6352.00	–	6339.00
EAGDE-SVM [46]	14.10	20.60	18.86	26.40	253.11	3122.20	514.50	4795.02
BiLSTM [47]	13.65	23.18	13.74	24.86	295.00	4130.00	317.00	5430.00
AdaBN-DCNN [48]	13.17	20.87	14.97	24.57	279.00	2020.00	817.00	3690.00
SMSAE [49]	12.99	–	–	22.31	270.74	–	–	3297.06
MSDCNN-LSTM [50]	12.96	18.70	11.78	21.57	256.59	1873.86	211.99	2699.34
STRUL [51]	12.85	19.24	13.74	22.34	224.00	1950.00	252.00	3080.00
ABGRU [52]	12.83	–	13.23	–	221.54	–	279.18	–
CATA-TCN [53]	12.80	17.61	13.16	21.04	234.31	1361.23	290.63	2303.42
PAOLTransformer [54]	12.49	21.63	12.66	23.86	257.71	1692.59	274.15	3163.41
AGCNN [55]	12.42	19.43	13.39	21.50	225.51	1492.00	227.09	3392.00
MSBLS [56]	–	16.39	–	17.75	–	1434.27	–	1785.33
Proposed SAM-SVAE	12.38	15.91	12.53	16.48	208.63	921.29	225.69	1636.99

Table 9

The network performance of each operational condition under different data availability.

Percentage	RMSE				S-score			
	FD001	FD002	FD003	FD004	FD001	FD002	FD003	FD004
2%	16.92	19.98	19.73	22.43	420.01	1524.05	1039.39	2758.54
5%	15.86	19.32	18.69	21.71	407.82	1488.47	976.70	2522.46
10%	15.70	19.02	16.68	21.46	401.59	1336.72	874.53	2403.48
30%	15.68	18.05	15.46	19.59	386.18	1209.29	799.71	2334.34
50%	13.34	17.28	14.73	18.53	301.72	1194.68	509.64	2086.09
100%	12.38	15.91	12.53	16.48	208.63	921.29	225.69	1636.99

**Fig. 7.** The calibration curves of FD001–FD004.

distribution. The PIT histograms of FD001–FD004 are shown in Fig. 8 in the paper. Under ideal calibration, the PIT values should follow a uniform distribution on the unit interval. For all four sub-datasets, the PIT histograms do not exhibit strong U-shaped or highly skewed patterns,

which would indicate severe under-dispersion or over-dispersion. Together, these results provide visual and quantitative evidence that the proposed SAM-SVAE produces well-calibrated predictive distributions for RUL across different operational conditions.

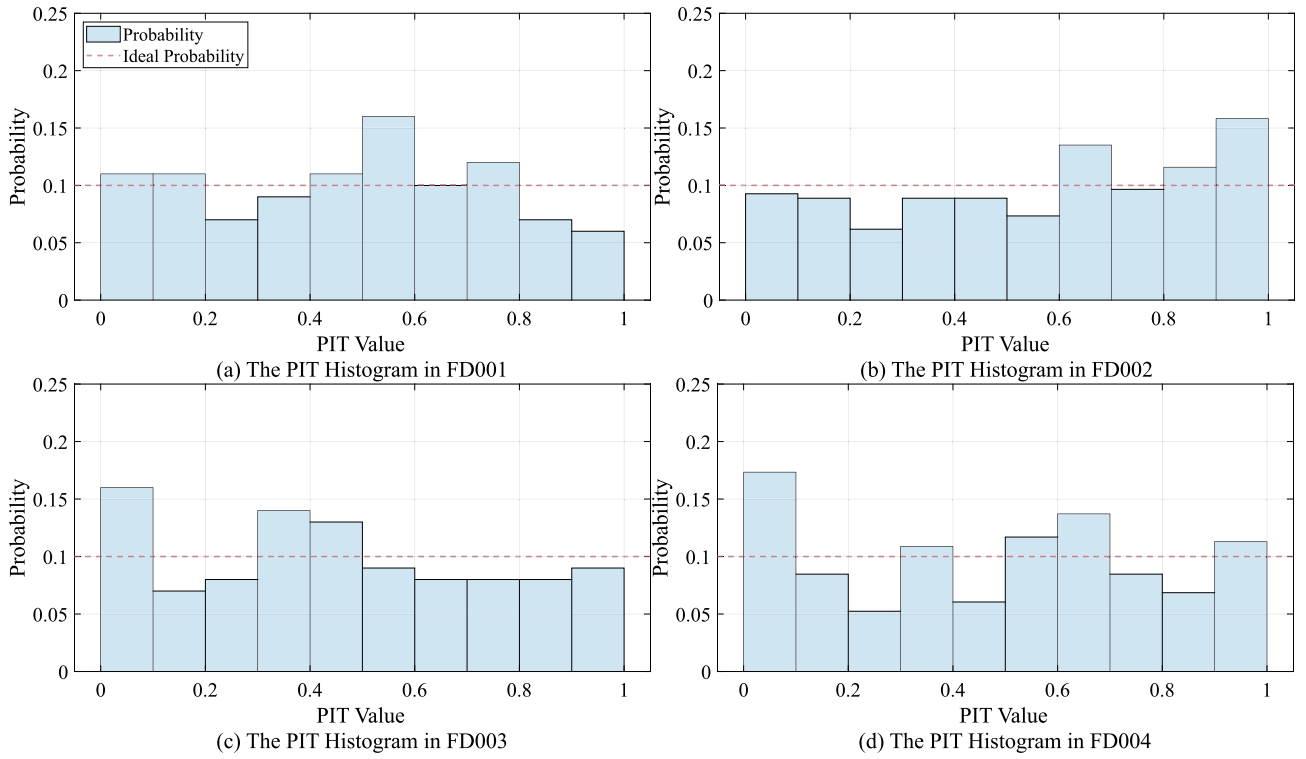


Fig. 8. The PIT histograms of FD001–FD004.

4.5. Network interpretability analysis

This paper investigates the determining impact of latent features on RUL prediction using semi-supervised SAM-SVAE, which focuses on the latent space visual analysis of the network. Considering the two dimensions of the mean and variance variables, a potential phase space trajectory can be constructed for the degradation process of each aircraft turbofan engine, which is presented in a scatter plot, showing the evolution over the time through a color gradient process. Specifically, this paper takes the aircraft turbofan engines numbered 24 and 76 under the FD001 (Fig. 9), the aircraft turbofan engines numbered 99 and 210 under the FD002 (Fig. 10), the aircraft turbofan engines numbered 82 and 99 under the FD003 (Fig. 11), the aircraft turbofan engines numbered 22 and 111 under the FD004 (Fig. 12). Sub-graphs (a) and (c) of Figs. 9–12 present a comparative analysis between the actual RUL values and model prediction. In addition, the corresponding evolution of the latent variables in the degradation is shown in sub-graphs (b, d) in Figs. 9–12.

As shown in Figs. 9–12, unlike conventional data-driven RUL prediction approaches, which typically provide only point estimates, the proposed approach yields both predicted RUL values and corresponding confidence intervals (demonstrated with a 95% confidence level in this study), effectively quantifying prediction uncertainty. During the early operational phase of an aircraft turbofan engine, the latent variable trajectories exhibit relatively flat trends with minimal variance, indicative of stable operational characteristics. This stage corresponds to early-stage physical degradation processes, such as microscopic material fatigue and minor surface wear, which have negligible immediate effects on engine performance and reflect gradual deterioration. As operation progresses into intermediate and later stages, the latent variable trajectories manifest distinctly nonlinear accelerated trends, accompanied by increased variance and greater dispersion, signaling heightened system state uncertainty. This corresponds mechanically to rapid internal physical degradation phenomena, including accelerated material fatigue crack propagation, intensified bearing raceway wear,

and abnormal blade vibration frequencies. These observable changes in latent variable trajectories indicate the evolving physical degradation conditions. From the perspective of uncertainty quantification, the variance changes within the latent variable space clearly illustrate the evolution of prediction uncertainty correlated with equipment degradation. Specifically, during early operational stages, smaller variance reflects higher model confidence in stable condition predictions. Conversely, continued degradation leads to progressively larger variances and increased dispersion, emphasizing rising uncertainty in model predictions and highlighting the necessity for targeted monitoring or maintenance interventions. This quantification of uncertainty, as depicted through latent variable variance and dispersion trends, notably enhances the interpretability of model decision-making processes. Additionally, this analysis provides an interpretability channel for the proposed approach. Over time, the predictive accuracy of the deep neural network progressively improves, consistent with the RUL prediction comparison results presented in Tables 7–9.

5. Conclusion

To make full use of unlabeled degradation data and focus on key parts in the time window through serialization, this paper proposes an SAM-SVAE for RUL prediction from a semi-supervised perspective. Specifically, this paper designs an attention network to focus on key parts of the time window. Furthermore, three probabilistic deep generation networks are constructed to describe the hidden variable space, from which an SAM-SVAE is designed for making full use of unlabeled degradation data to improve the performance. The principal contributions are outlined as follows: (1) A self-attention mechanism with dynamic serialization module, coupled with a BiGRU encoder is proposed. The module concentrates modeling capacity on salient time steps, while preserving long-short range dependencies in multivariate sensor streams. (2) An SAM-SVAE approach induces a learned hidden-variable state space that is explicitly aligned with a state-space model from control theory. The inference sub-network, encoding sub-network, and decoding sub-network jointly define a full predictive RUL

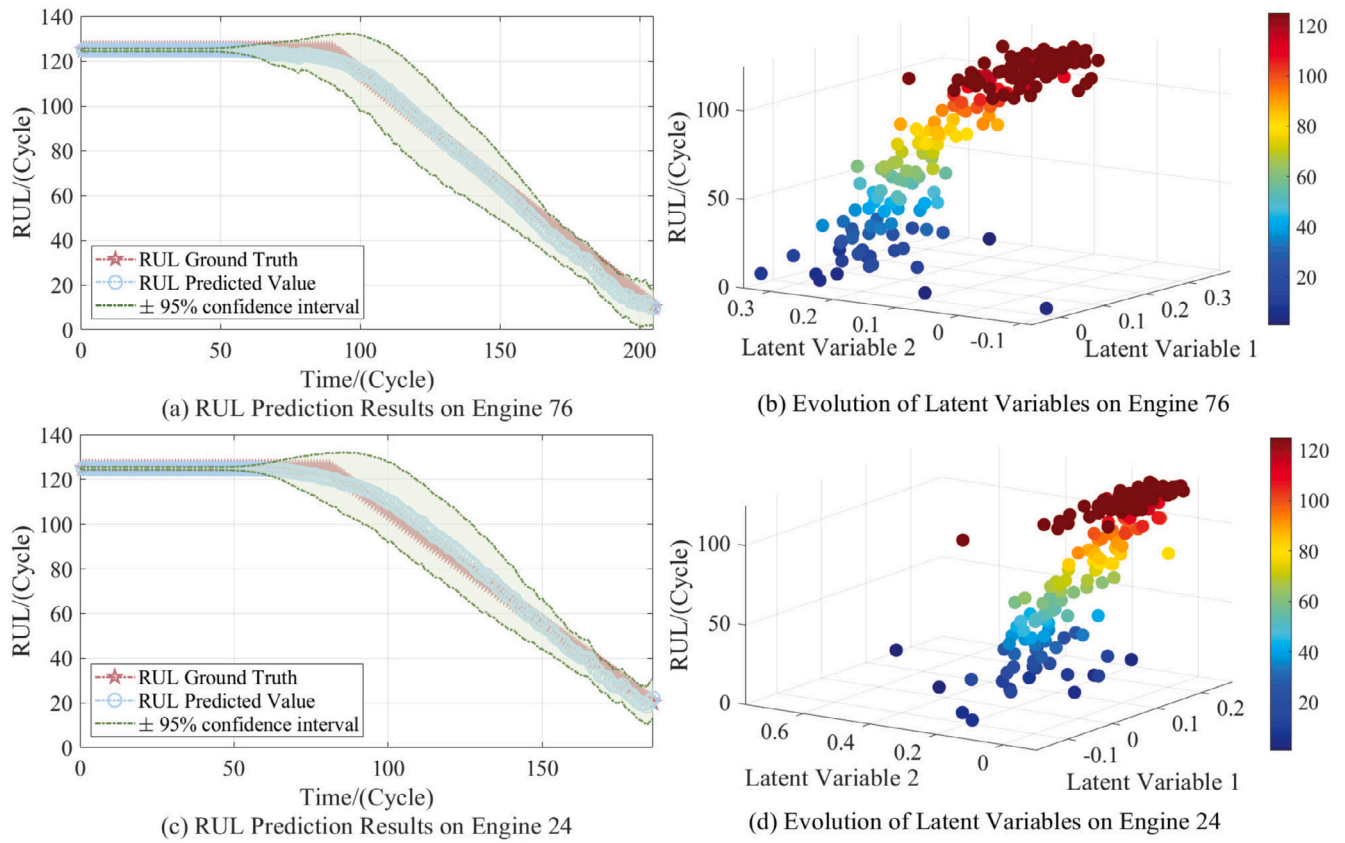


Fig. 9. The evolution of the latent variables and RUL predicted value in the degradation process under FD001.

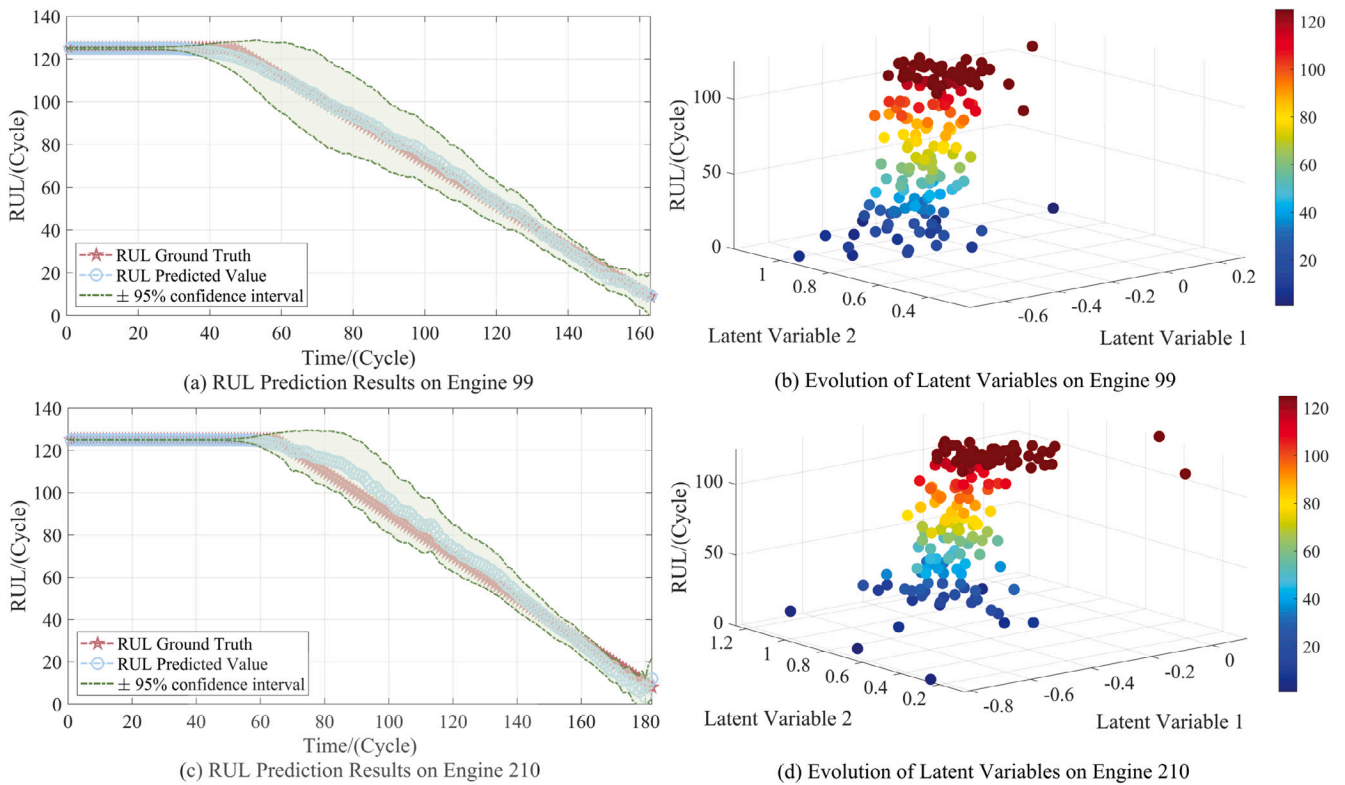


Fig. 10. The evolution of the latent variables and RUL predicted value in the degradation process under FD002.

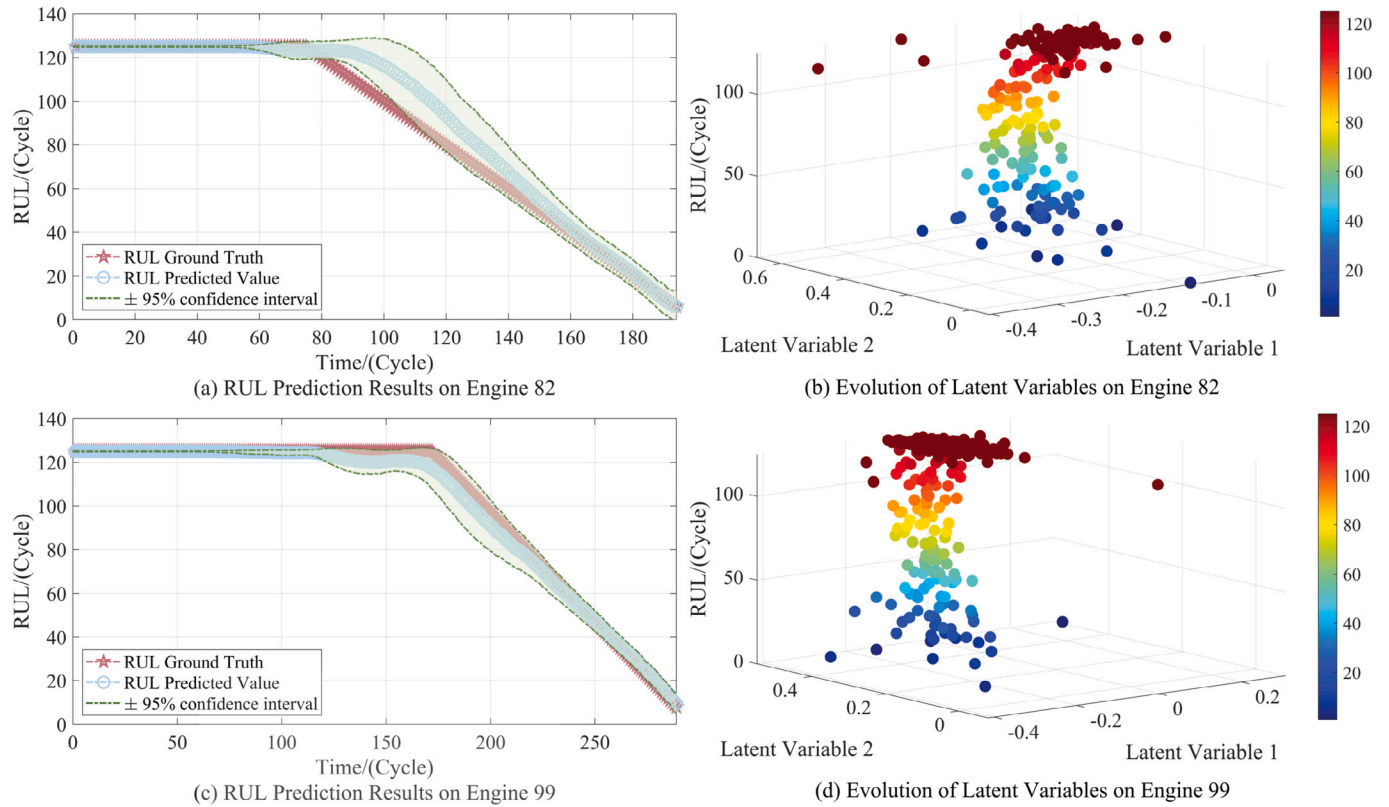


Fig. 11. The evolution of the latent variables and RUL predicted value in the degradation process under FD003.

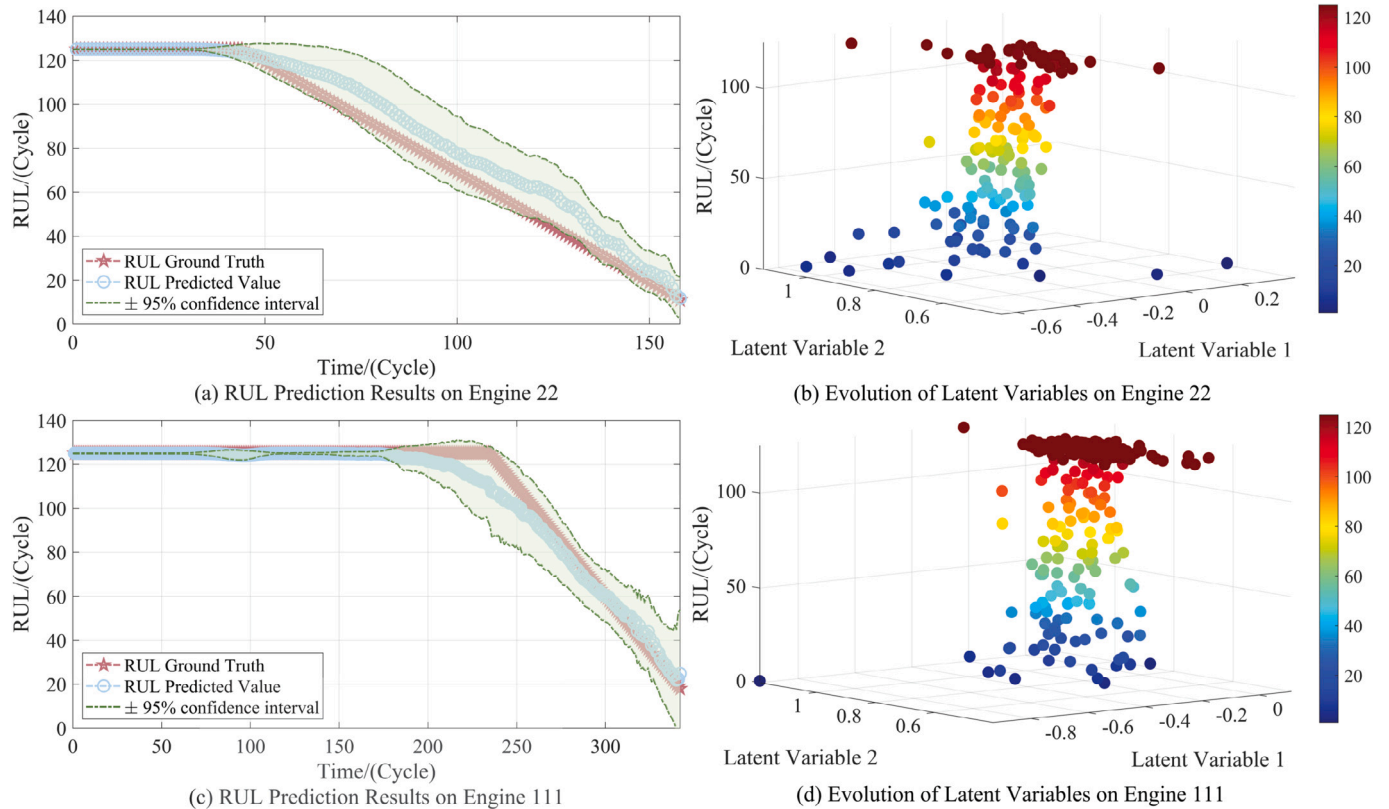


Fig. 12. The evolution of the latent variables and RUL predicted value in the degradation process under FD004.

distribution, enabling calibrated uncertainty quantification and providing an interpretable latent trajectory consistent with control-theoretic dynamics. (3) A semi-supervised training objective jointly optimizes a supervised likelihood with an unsupervised feature reconstruction, which allows effective use of large unlabeled degradation datasets without synthetic or pseudo-labels.

CRedit authorship contribution statement

Jiushi Zhang: Writing – original draft, Validation, Methodology, Investigation, Funding acquisition, Conceptualization. **Kai Chen:** Writing – review & editing, Supervision. **Fan Wu:** Methodology, Conceptualization. **Quan Qian:** Writing – review & editing, Methodology, Conceptualization. **Tenglong Huang:** Writing – review & editing, Methodology, Conceptualization. **Yuhua Cheng:** Writing – review & editing, Supervision. **Shen Yin:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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